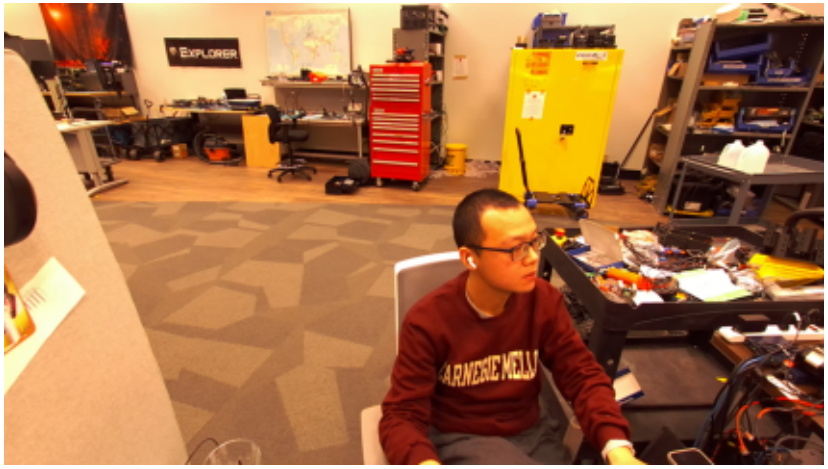
ZED depth: ONNX vs PyTorch

PASS: outputs within tolerance

Model: neural_depth_5.3.onnx | atol=0.003, rtol=0.003

Left RGB input: im0.png



Right RGB input: im1.png



Output	Mean abs.	Max abs.	RMSE	P99 abs.	Outside tol.	Result
disparity	4.83699e-06	0.00107956	1.89421e-05	6.10352e-05	0%	PASS
disparity_right	0	0	0	0	0%	PASS
confidence	5.10937e-07	5.57899e-05	1.62939e-06	7.27177e-06	0%	PASS

Input HxW: [621, 1104] | network HxW: [640, 1152] | bottom/right padding: [19, 48]

Output shape: [1, 1, 320, 576] | comparison includes padding; disparity is not metric depth.

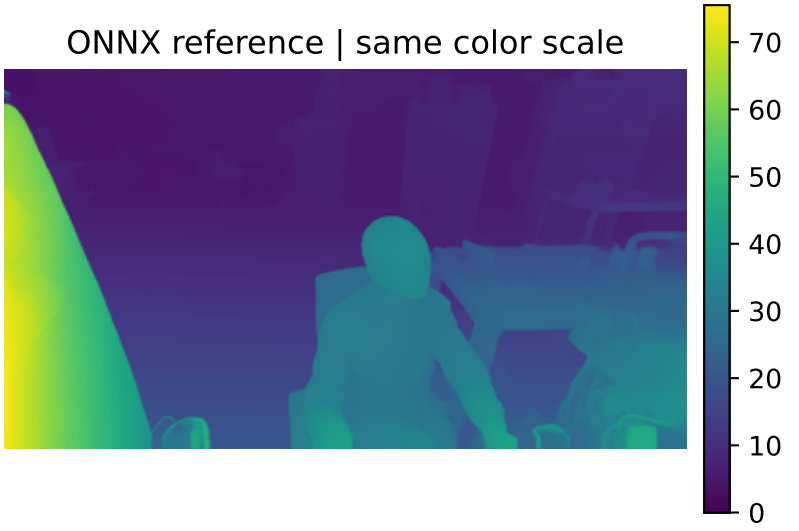
Reference: ONNX Runtime standard layers + independent NumPy custom operators. No PyTorch tensors are injected.

disparity_right is a zero-filled placeholder; matching zeros does not validate right-view depth.

This measures implementation agreement on one stereo pair. No ground truth or native ZED CUDA parity was tested.

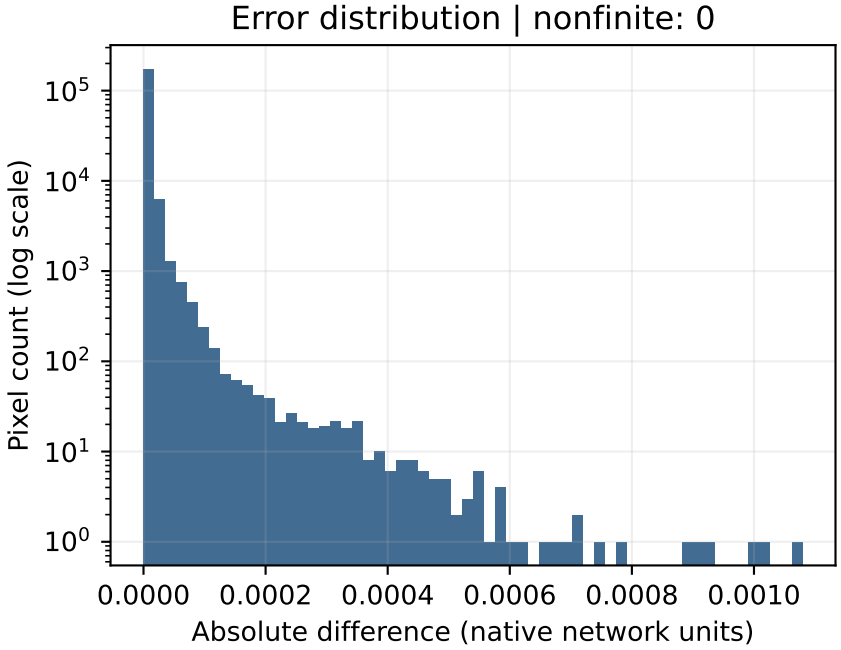
Timing: PyTorch forward 0.728s; reference setup + run 1.030s.

Disparity output analysis



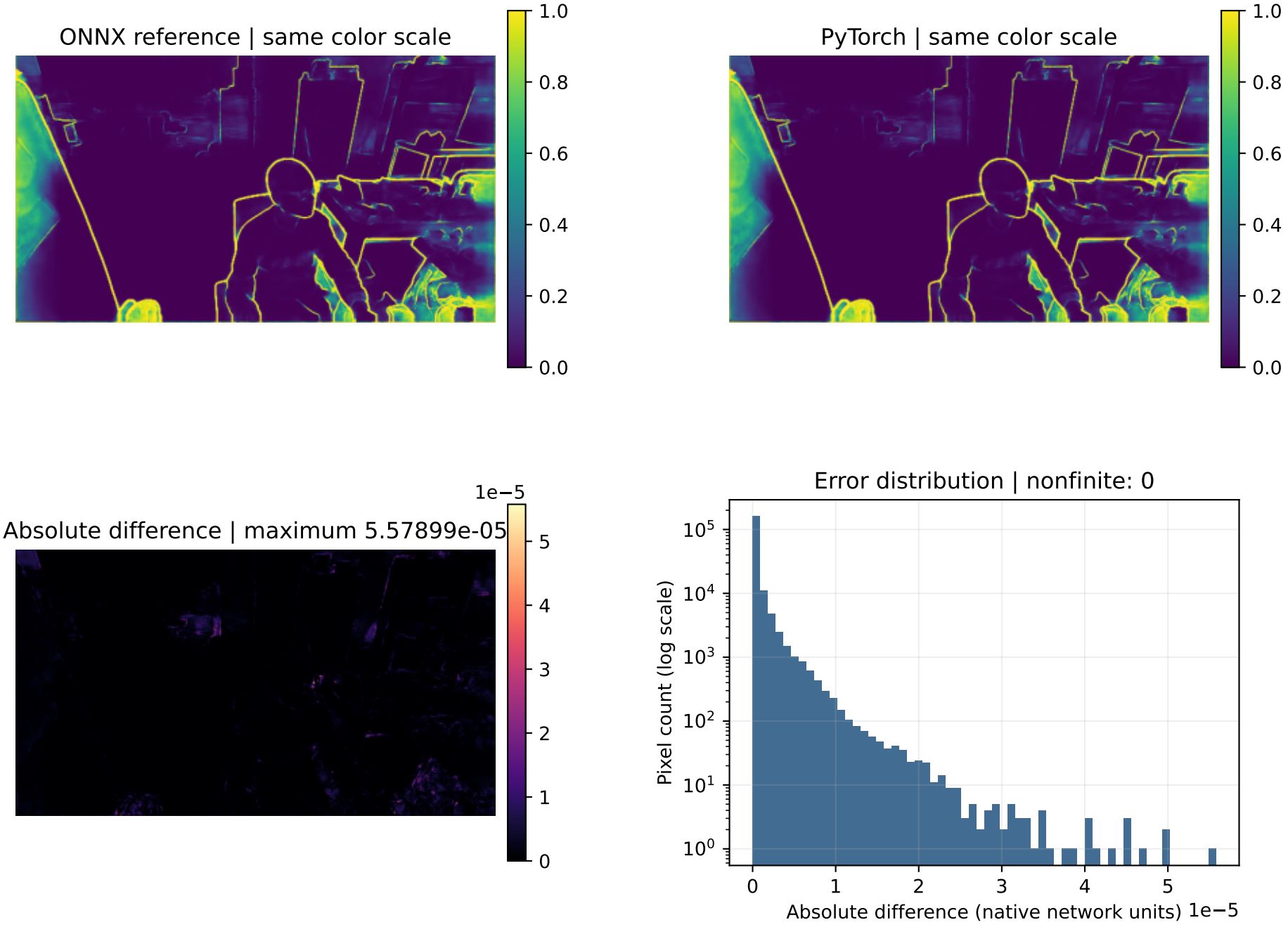






Out-of-tolerance: 0% | Difference colors are magnified to the maximum error; inspect numeric values, not color alone.

Confidence output analysis



Out-of-tolerance: 0% | Difference colors are magnified to the maximum error; inspect numeric values, not color alone.